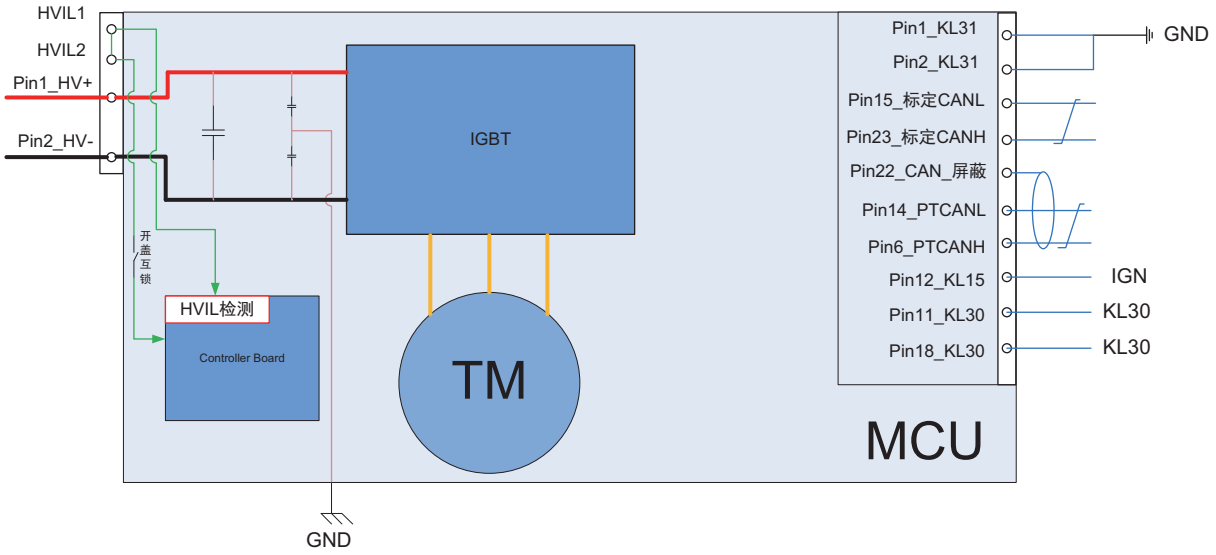
P0AEF-11、 P0AF4-11、 P0BD3-11、 P0AF0-12、 P0AF5-12、 P0BD4-12

Fault diagnosis code
P0AEF-11：U-phase IGBT temperature signal short-circuit or open circuit to ground
P0AF4-11：V-phase IGBT temperature signal is short-circuited or broken to ground
P0BD3-11：W-phase IGBT temperature signal is short-circuited or broken to ground
P0AF0-12：U-phase IGBT temperature signal shorted to the power supply
P0AF5-12：V-phase IGBT temperature signal shorted to the power supply
P0BD4-12：W-phase IGBT temperature signal shorted to the power supply
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Cooling system fault • The IGBT of PEU fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0AEF-11
AD value is less than 1.6%,and the difference between U-phase IGBT temperature and motor temperature is greater than 20 ℃ , all 3 phases fail and reduce the power. Otherwise, do not deal with it.
P0AF4-11
AD value is less than 1.6%,and the difference between V-phase IGBT temperature and motor temperature is greater than 20 ℃ , all 3 phases fail and reduce the power. Otherwise, do not deal with it.
P0BD3-11
AD value is less than 1.6%,and the difference between W-phase IGBT temperature and motor temperature is greater than 20 ℃ , all 3 phases fail and reduce the power. Otherwise, do not deal with it.
P0AF0-12
AD value is greater than 93%, and all 3 phases fail and reduce the power. Otherwise, do not deal with it.
P0AF5-12
AD value is greater than 93%, and all 3 phases fail and reduce the power. Otherwise, do not deal with it.
P0AF4-12
AD value is greater than 93%, and all 3 phases fail and reduce the power. Otherwise, do not deal with it.
To set the effect of a fault code condition
No obvious feeling

Circuit diagram



EV901001

Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 If there is any fault code except for P0AEF-11、 P0AF4-11、 P0BD3-11、 P0AF0-12、 P0AF5-12、 P0BD4-12?
	YES → Refer to:DTC Summary list(PEU).
NO ↓	
Step 3	Check cooling system.
	1 Check whether there is leakage or poor heat dissipation in the cooling system. 2 Check whether the result is normal or not?
	NO → Repair the cooling system.
YES ↓	
Step 4	Check the connecting cable of PEU.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of PEU. 5 Check whether the PEU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized with the diagnostic instrument. 5 Can the governor be initialized and is the governor restored?
		NO → Repair or change the governor.
YES ↓		
Step 6	Check PEU internal circuit.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check the PEU internal circuit and components for obvious damage or ablation. 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 7	Test whether PEU is norm.	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
		YES → Change the PEU Module
NO ↓		
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.		
Next Step ↓		

Diagnosis end.